GENERAL PROJECT GUIDELINES

1. General Context

As part of the *Oracle Database Administration* course, you are required to complete an engineering project inspired by real-world challenges faced by companies.

These projects aim to develop your professional skills in Oracle database administration, database architecture, automation, cloud computing, DevOps, cybersecurity, monitoring, performance optimization, and IT service continuity.

You are a team of two engineering consultants hired by a company to solve a specific IT problem.

You must approach your project as a real professional consulting and engineering assignment.

This is not simply an academic exercise involving software installation, running a few commands, or following an online tutorial.

You are expected to analyze a business problem, propose an appropriate solution, design its architecture, develop a working implementation, and demonstrate its effectiveness through measurable results.

Your main objective is to deliver a solution that meets your client's requirements and the specifications provided in your project assignment.

2. Professional Scenario

For each project, you will work with a client company operating in a sector such as banking, insurance, telecommunications, e-commerce, manufacturing, healthcare, logistics, or IT services.

You will respond to a request from an IT manager, such as a Chief Information Officer (CIO), Infrastructure Manager, DBA Manager, Security Manager, or DevOps Manager.

Your project specifications will include:

- **Client Company:** Its business sector, IT environment, and professional context.
- **Project Sponsor:** The IT manager who assigns the project to your team.
- **Business Problem:** The specific problem you are expected to solve.
- **Assigned Mission:** The services, responsibilities, and expected outcomes.
- **Operational Constraints:** Availability, security, performance, budget, deadlines, compatibility, and resources.
- **Expected Results:** Measurable indicators and acceptance criteria used to evaluate your solution.

You must adopt a professional approach and justify your decisions as you would when working with a real client.

Your role is not limited to performing technical tasks. You must also analyze risks, propose solutions, justify your technical choices, and present your results professionally.

3. Your Team's Responsibilities

You are responsible for managing the entire lifecycle of your engineering assignment.

You must:

1. Analyze the business problem presented by your client.
2. Identify functional and technical requirements.
3. Investigate possible technological solutions.
4. Propose a suitable architecture and justify your choices.
5. Plan your activities and distribute responsibilities within your team.
6. Install and configure the required technical environment.
7. Develop the necessary scripts, programs, and automation mechanisms.
8. Implement the required functionalities.
9. Conduct appropriate functional, technical, security, or performance tests.
10. Measure, analyze, and interpret the results.
11. Identify technical difficulties and propose solutions.
12. Evaluate the limitations of your implementation and suggest improvements.
13. Prepare the required deliverables and present the results of your mission.

Both team members must actively participate in the design, implementation, and validation of the solution.

You must document how responsibilities are distributed within your team.

Each team member must be able to explain the overall architecture, the main scripts, and how the solution works.

4. General Technical Requirements

4.1 Architecture and Design

You must design a clear technical architecture that addresses the client's problem and is appropriate for the available resources.

Your architecture must identify:

- Hardware and software components.
- The role of each component.
- Interactions between the different services.
- Data and communication flows.
- Security mechanisms.
- Technical dependencies.
- Design choices supporting reliability and maintainability.

You must provide an **overall system architecture diagram** and any additional diagrams necessary to explain your solution.

Depending on your project, these may include:

- Technical architecture diagram.
- Deployment diagram.

- Sequence diagram.
- Activity diagram.
- Component diagram.
- Data flow diagram.
- Conceptual or logical data model, when relevant.

You should select only the diagrams that are relevant to your project and explain their purpose.

4.2 Functional Implementation

You must develop a working technical solution that meets the main requirements of your project specifications.

Your implementation may take the form of a platform, infrastructure, automated system, administration tool, monitoring solution, or security mechanism.

You must implement the main technical components yourselves and be able to demonstrate how they work.

Simply following a tutorial or installing a technology without meaningful integration and testing is not sufficient.

4.3 Automation and Code Quality

Whenever appropriate, you must prioritize the automation of repetitive tasks.

You may use SQL, PL/SQL, Python, Bash, Java, Ansible, Terraform, GitLab CI, Jenkins, Docker, or any other suitable technology.

Your scripts and programs must be:

- Properly structured.
- Readable and documented.
- Organized into appropriate directories.
- Version-controlled using Git.
- Tested and functional.
- Reusable whenever possible.
- Free from hardcoded passwords, credentials, and other sensitive information.

You must be able to explain the purpose and operation of your main scripts.

4.4 Security and Compliance

You must apply security best practices appropriate to your project.

In particular, you must:

- Apply the principle of least privilege.
- Properly manage user accounts and access permissions.
- Protect sensitive information.
- Separate environments whenever necessary.
- Log important operations.

- Secure credentials and connection parameters.
- Comply with Oracle licensing requirements and the terms of use of other software.
- Use only test data, anonymized data, or data you are authorized to process.

You must also identify the main security risks associated with your solution and explain the protection measures you have implemented.

4.5 Measurable Results

Before conducting your tests, you must define measurable indicators that allow you to evaluate your solution objectively.

Depending on your project, these indicators may include:

- Response time and latency.
- Transaction throughput.
- Service availability.
- Recovery Time Objective (RTO).
- Recovery Point Objective (RPO).
- Mean Time to Detect (MTTD).
- Mean Time to Repair or Restore (MTTR).
- Test success rate.
- Deployment time.
- Reduction in manual operations.
- CPU, memory, and storage consumption.
- Number of detected errors.
- Compliance rate with security controls.

You must present the actual results obtained and, whenever possible, compare performance before and after implementing your solution.

Your findings must be supported by evidence such as logs, commands, screenshots, measurement tables, graphs, or test reports.

Statements such as "our solution is efficient, reliable, and secure" must be supported by verifiable technical evidence.

5. Project Organization and Schedule

You have four weeks to complete your assignment.

You must organize your work according to the following schedule:

WEEK	ACTIVITIES	EXPECTED OUTCOMES
WEEK 1	Requirements analysis, solution assessment, architecture design, technology selection, and planning	Validated design and prepared environment
WEEK 2	Installation, configuration, development, and implementation	First working version

WEEK 3	Integration, testing, incident simulation, optimization, and performance measurements	Verifiable technical results
WEEK 4	Final improvements, documentation, deliverable preparation, and project defense	Final operational solution and demonstration

You must keep a record of the main stages of your work, technical decisions, problems encountered, and corrective actions.

6. Required Deliverables

At the end of your assignment, you must submit **five mandatory deliverables**.

Deliverable 1 — Complete Technical Report (PDF)

You must prepare a professional technical report of **25 to 40 pages, excluding appendices**, presenting your entire engineering approach.

Your report must include the following sections:

1. Cover Page

Include your institution, Master's program, course name, project title, team members, supervising professor, and academic year.

2. Abstract and Keywords

Briefly describe the problem, assigned mission, implemented solution, and main results.

3. General Introduction

Present the project context, motivation, problem statement, and objectives.

4. Client Company Presentation

Describe the business sector, IT environment, project sponsor, business requirements, and operational constraints.

5. Requirements Analysis and Project Specifications

Identify functional and technical requirements, constraints, and acceptance criteria.

6. Technology Assessment and Justification

Present the possible technical solutions, compare them, and justify your choices.

7. System Design and Architecture

Provide the overall system architecture diagram, relevant technical diagrams, system components, interactions, and data flows.

8. Implementation Environment

Describe the hardware, operating systems, software, versions, configurations, and prerequisites.

9. Solution Implementation

Explain the implementation steps, configurations, developed mechanisms, and main scripts used.

10. Practical Usage Examples

Present several practical scenarios supported by commands, screenshots, code examples, and explanations of the results.

11. Testing and Validation

Describe your testing methodology, executed scenarios, results, and verification of acceptance criteria.

12. Performance and Reliability Evaluation

Present measured indicators, results, comparison tables, graphs, and their interpretation.

13. Challenges Encountered and Solutions

Explain the main technical problems, troubleshooting approaches, and solutions implemented.

14. Discussion and Future Improvements

Evaluate the strengths, limitations, remaining risks, and possible improvements of your solution.

15. General Conclusion

Summarize the outcome of your mission and explain to what extent the client's objectives have been achieved.

16. References and Appendices

Include official documentation, technical references, and any additional supporting material.

Report requirements:

- Use clear, academic, and professional English.
- Explain your architectural decisions and technical choices.
- Comment on your results and screenshots.
- Include real usage examples from your implementation.
- Provide readable diagrams that accurately represent your solution.
- Avoid filling the report with unexplained screenshots.
- Store long scripts in your Git repository and include only relevant code excerpts in the report.

Deliverable 2 — Separate Project Defense Presentation

You must prepare a **PowerPoint (PPTX) or PDF presentation containing 12 to 15 slides**.

Your presentation must cover:

1. Company background and project context.
2. Business problem and client requirements.
3. Assigned mission and objectives.
4. Proposed technical architecture.
5. Selected technologies and justification.
6. Main implementation stages.
7. Implemented features and components.
8. Practical usage examples.
9. Testing and validation scenarios.
10. Measurable results.
11. Challenges encountered and solutions.
12. Limitations, recommendations, and conclusion.

You should prioritize meaningful diagrams, results, demonstrations, and technical illustrations.

Your presentation must summarize your mission rather than reproduce the full technical report.

Deliverable 3 — Source Code and Executable Scripts

You must submit all technical components developed or configured during your project.

Depending on your assignment, these may include:

- SQL and PL/SQL scripts.
- Python, Java, or other programming language source files.
- Bash and Shell scripts.
- Configuration files.
- Automation scripts.
- Terraform or Ansible configuration files.
- CI/CD pipeline configurations.
- Backup and recovery scripts.
- Monitoring tools and dashboards.
- Security and auditing scripts.
- Testing scripts.
- Sample datasets required for the demonstration.

You must organize your files and provide clear execution instructions.

Deliverable 4 — GitHub or GitLab Repository

You must provide a well-organized GitHub or GitLab repository containing your complete work.

The repository must include:

- A README.md file describing your project.
- Hardware and software prerequisites.
- Installation instructions.
- Required configuration procedures.
- Source code and scripts.
- Architecture diagrams and technical diagrams.
- Testing scenarios.
- Results and evidence of correct operation.
- Final technical report in PDF format.
- Project defense presentation.

Your repository must allow an external reviewer to understand your project and reproduce the solution under the documented conditions.

You must not publish passwords, access credentials, private keys, confidential data, or other sensitive information.

Deliverable 5 — Working Technical Demonstration

You must present a practical demonstration of your implemented solution.

Your demonstration must verify:

1. The operation of the main functionalities.
2. The correct execution of scripts and automation procedures.
3. At least one realistic business scenario.
4. The behavior of your solution when a relevant error or incident occurs.
5. The results and performance indicators obtained.

You must prepare your environment to ensure that your demonstration is consistent with the work and results presented in your technical report.

7. Project Defense and Assessment

You must present and defend your work before an evaluation committee.

The project defense will last **15 minutes per team**, organized as follows:

ACTIVITY	DURATION
PRESENTATION OF THE CONTEXT, SOLUTION, AND ARCHITECTURE	3 minutes
LIVE TECHNICAL DEMONSTRATION	5 minutes
PRESENTATION OF TESTS AND RESULTS	2 minutes
QUESTIONS FROM THE COMMITTEE AND INDIVIDUAL ASSESSMENT	5 minutes
TOTAL DURATION	**15 minutes**

Both team members must be able to answer technical questions related to the architecture, implementation, scripts, tests, technology choices, and results.

Your individual understanding of the work will be taken into account during the evaluation.

8. Academic Integrity and Use of Artificial Intelligence

You may consult official documentation, technical resources, professional communities, and artificial intelligence tools to support your learning and project development.

However, you must:

- Understand and explain the scripts and configurations you use.
- Verify that the proposed solutions work correctly.
- Test and validate any generated or adapted code.
- Acknowledge the main resources consulted.
- Demonstrate your own contributions to the analysis, design, and implementation.
- Avoid reproducing existing projects without meaningful technical contributions.
- Be able to explain, modify, or correct your solution during the project defense.

Using artificial intelligence tools does not remove your responsibility to understand the technologies, concepts, implementations, and results presented.

9. Project Success Criteria

Your mission will be considered successfully completed when:

1. You have understood and clearly explained the business problem.

2. You have proposed a coherent and technically justified architecture.
3. You have developed a working solution that meets the main requirements.
4. You have demonstrated the required functionalities.
5. You have conducted reproducible and documented tests.
6. You have presented concrete and measurable results.
7. You have analyzed incidents, technical challenges, and limitations.
8. You have submitted all required deliverables.
9. You have organized your repository and documented the installation and usage procedures.
10. Both team members can confidently defend the technical decisions before the evaluation committee.

10. Final Instructions — Your Professional Commitment

You are a team of engineering consultants hired by a company to solve a real-world IT problem.

You must maintain a professional engineering approach throughout your assignment.

Your responsibility is to understand your client's requirements, design an appropriate architecture, develop a working solution, verify its operation, and demonstrate its effectiveness.

You must also identify the limitations of your implementation and provide relevant recommendations for improvement.

Your final objective is not simply to deliver a working program or infrastructure, but to demonstrate your ability to successfully carry out a professional Oracle Database Administration and Engineering assignment.

At the end of your project, you should be able to present your work to a Chief Information Officer, DBA Manager, or DevOps Manager as a professional, documented, reproducible, and technically justified solution.